

Michael H. Gross

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SUMMARY

Robotics software engineer spanning firmware and hardware interfaces through real-time control to concurrent production systems.

EXPERIENCE

Robotics Engineer

Jan — Sep 2026

Workr (NVIDIA Inception Member), Mountain View, CA

- Architected the concurrent runtime as a multi-process, event-driven system over a custom Protobuf IPC transport and thread-safe MPSC event queue, each process governed by its own finite state machine (FSM), fault-isolated and independently restartable
- Built a lock-free real-time control loop above 1 kHz, streaming non-deterministic paths at sub-0.5 mm per-step positional accuracy
- Integrated GPU-accelerated trajectory optimization with dynamic collision-aware volumetric scene generation; benchmarked ~100 ms faster than ABB's per-path native planner and lowered cycle times by 44% through continuous path generation and chaining
- Led the Temporal app orchestration layer coordinating closed-loop control, motion planning, and the robot control interface; authored 46% of a greenfield Python production codebase spanning 18 sub-packages and 8 independently deployed microservices
- Deployed and operated the production stack as sole owner at 99.7% uptime, building the remote debugging and developer tooling that cut on-site production fault diagnosis from hours to minutes; delivered 6 customer and investor demonstrations end-to-end

Graduate Research Assistant

Aug 2024 — Aug 2025

ASU Robotics and Autonomous Systems Lab, Mesa, AZ

- Developed [armctl](#), a vendor-agnostic Python API for UR, Vention, and Elephant Robotics arms; 15 ms round-trip, 5k+ downloads
- Created [mujoco-toolbox](#), a MuJoCo-based digital-twin simulation toolkit with XML/URDF merging, inverse kinematics, and trajectory generation; 9k+ PyPI downloads, supporting reinforcement learning and reproducible closed-loop robot experiments
- Containerized CI/CD workflows that standardized benchmarking and eliminated hours of per-machine configuration overhead

Research & Development Intern

May — Aug 2024

Precision Planting (An AGCO Company), Fargo, ND

- Developed tractor heading-modulo data-cleaning algorithms for optical seeder sensor field data across 10k+ acres of pilot farmer fields; analysis showed 10% lower detection variance than competitor products, directly feeding an in-house production pipeline
- Integrated embedded sensors into existing CAN bus infrastructure with custom drivers, enabling future product-line development
- Designed lab test fixtures, including a 6-DOF Stewart platform, replicating extreme field motion to validate developmental sensors

EDUCATION

Arizona State University, Mesa, AZ

Aug 2025

Robotics and Autonomous Systems, M.S.

with Distinction — 4.0 GPA

Robotics Engineering, B.S.E.; Minor in Applied Mathematics

Summa Cum Laude — 4.0 GPA

PROJECTS

[Robotic Glovebox Digital Twin](#) | Los Alamos National Laboratory | M.S. Thesis

Aug 2024 — Aug 2025

- Built robotic arm digital twin for hazardous, confined glovebox workcells with computer vision, motion planning, and control
- Applied deep learning detection with fiducial marker sensor fusion for robust tracking under occlusion and limited visibility

[Parkinson's Rehabilitation Device](#) | Barrow Neurological Institute

Jan — May 2025

- Engineered wearable medical robotics system with IMU sensors and BLE streaming; sub-50 ms sensor-to-actuator feedback loop for vibrotactile therapy. Developed embedded firmware and a responsive client-side rehabilitation application built on tk-engine

[Turtlebot4 Predator-Prey Model](#) | Arizona State University

Jan — May 2025

- Built ROS2 nodes for SLAM, computer vision detection, sensor fusion, and localization to enable autonomous navigation
- Designed and tested policy models and motion planning for multi-agent pursuit-evasion in cluttered, dynamic environments

[Frog Gait Reproduction](#) | Arizona State University

Aug — Dec 2024

- Modeled five-bar limb kinematics and optimized material parameters in MuJoCo to reproduce gait in a foldable bio-inspired robot

[Tilt-Up Window Automation](#) | Summit Automation

Aug 2023 — May 2024

- Developed 400 lb IoT device with linear actuators, embedded control, voice recognition, and custom PCB; delivered commercially

LEADERSHIP

Head Programming Mentor, FIRST Tech Challenge — Pioneer Robotics, Tucson, AZ

Aug 2022 — Present

Teach Kalman-filter sensor fusion, fiducial detection, localization, and path planning across a 20+ person club; founding member

SKILLS

Languages: Python, Rust, C/C++, MATLAB, Kotlin

Frameworks & Tools: ROS2, MuJoCo, Temporal, Protobuf, Docker, CI/CD, Git, Linux (Ubuntu, NixOS), CAN, SPI, I2C, UART

CAD & Fabrication: SolidWorks, Fusion 360, PCB Design (KiCad), 3D Printing, CNC Machining, Laser Cutting, Soldering